# Augmented Reality Solar System using Unity & Vuforia

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## 1. Project Overview

This project is an **Augmented Reality (AR) application** that visualizes the solar system using Unity and Vuforia.
Users can scan image targets to display 3D planets, interact with them, and view detailed information in a modern UI.

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## 2. Technologies Used

- Unity Engine
- Vuforia Engine
- C# Programming
- Unity UI System
- Android Build Support

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## 3. Project Features

- AR-based planet visualization
- Interactive UI (Start & Sound control)
- Clickable planets
- Info panel with 2D infographic images
- Smooth planet rotation
- Auto-hide UI when target is lost

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# 4. Project Structure

### Hierarchy

```
Canvas
 ├── MainUI
 │     ├── Background
 │     ├── StartButton
 │     └── SoundButton
 │
 └── InfoPanel
       └── PlanetImage

GameManager
AudioManager
AR Camera
ImageTargets (Planets)
Directional Light
EventSystem
```

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# 5. Implementation Steps

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### Step 1: Create Unity Project

- Create a new 3D Unity project
- Import Vuforia Engine

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### Step 2: Setup AR Environment

- Enable Vuforia in Player Settings
- Add AR Camera
- Add Image Targets for planets

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### Step 3: Add 3D Planet Models

- Import planet models
- Attach each model to its corresponding Image Target

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### Step 4: Create UI

- Create Canvas (Screen Space - Overlay)
- Add:

    - Background image
    - Start Button
    - Sound Button
    - Info Panel

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## Step 5: Info Panel Design

- Panel with semi-transparent background
- Positioned on screen side
- Contains:
    - Planet image (2D infographic)

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## Step 6: Add Interactivity

### Planet Click Script

- Detect user touch/click using Raycast
- Display corresponding planet image

### Info Controller

- Show/Hide panel
- Toggle content dynamically

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## Step 7: Smooth Rotation

- Apply smooth rotation using Quaternion interpolation
- Improves visual stability in AR

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## Step 8: Target Tracking Handling

- Detect when Image Target is lost
- Automatically hide info panel

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## Step 9: Audio System

- Background music using Audio Source
- Toggle sound using UI button

### Step 10: Build for Android

- Set platform to Android
- Set Minimum API Level (29+)
- Build APK

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## 6. Scripts Used

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### StartAR.cs

Handles UI start logic

```
public void StartApp()
{
    mainUI.SetActive(false);
    infoController.HideCompletely();
}
```

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### PlanetClick.cs

Handles user interaction with planets

```
if (Physics.Raycast(ray, out hit))
{
    if (hit.transform == transform)
    {
        controller.ToggleInfo(planetImage);
    }
}
```

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### PlanetInfoController.cs

Controls info panel display

```
public void ToggleInfo(Sprite sprite)
{
    panel.SetActive(true);
    image.sprite = sprite;
}
```

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### SmoothRotate.cs

Provides smooth rotation

```
transform.rotation = Quaternion.Slerp(
    transform.rotation,
    targetRotation,
    smoothFactor * Time.deltaTime
);
```

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**AudioManager.cs**

Controls sound

```
audioSource.mute = isMuted;
```

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# 7. Challenges Faced

- Unity build errors (NDK, API level)
- Input System conflict
- Audio import issues
- AR tracking instability

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# 8. Solutions

- Switched Input Handling to "Both"
- Fixed Android SDK/NDK settings
- Reimported assets
- Used smoothing techniques for rotation

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# 9. Final Result

- Fully interactive AR solar system
- Smooth user experience
- Clean UI design
- Ready for mobile deployment

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# 10. Future Improvements

- Add animations (fade/zoom)
- Add voice narration
- Add planet comparison feature
- Improve UI with advanced effects

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# 11. References


- Unity Documentation
- Vuforia Documentation
- Vuforia Engine Tracking rate
- Online tutorials and forums
- unity asset store (3D Models)
- Flaticon (UI Icons)
- Pixabay (background sound)
- All Images generated by advanced prompt with nano banana


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# 12. Image Tracking rate (rated by Vuforia Engine)

| Image | Target Name | Type | Rating | Status | Date Modified |
|---|---|---|---|---|---|
| | 1775039295579 | Image | ★★★★★ | Active | Apr 01, 2026 |
| | 1775039265924 | Image | ★★★★★ | Active | Apr 01, 2026 |
| | 1775036340405 | Image | ★★★★★ | Active | Apr 01, 2026 |
| | 1775036446197 | Image | ★★★★☆ | Active | Apr 01, 2026 |
| | 1775036473287 | Image | ★★★★☆ | Active | Apr 01, 2026 |
| | 1775036500073 | Image | ★★★★★ | Active | Apr 01, 2026 |
| | 1775036427933 | Image | ★★★★☆ | Active | Apr 01, 2026 |
| | 1775036364270 | Image | ★★★★☆ | Active | Apr 01, 2026 |
| | 1775036628404 | Image | ★★★☆☆ | Active | Apr 01, 2026 |

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# Conclusion

This project demonstrates how AR can be used to create engaging and educational applications.
By combining Unity and Vuforia, we successfully built an interactive solar system experience with real-time tracking and user interaction.

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